

Significant role of Bioactive compounds of *Ganoderma lucidum* in the aging process

Newton Paul *

Department of Zoology, Isabella Thoburn College, Lucknow-226007 Uttar Pradesh, India.

Open Access Research Journal of Biology and Pharmacy, 2025, 14(02), 035-044

Publication history: Received on 11 July 2025; revised on 20 August 2025; accepted on 22 August 2025

Article DOI: <https://doi.org/10.53022/oarjbp.2025.14.2.0033>

Abstract

Ganoderma lucidum, commonly known as Reishi or Lingzhi, is a medicinal mushroom widely recognized for its therapeutic potential in promoting health and longevity. Aging is a multifactorial process driven by oxidative stress, chronic inflammation, mitochondrial dysfunction, and immune decline. Bioactive compounds present in *G. lucidum*, such as polysaccharides (particularly β -D-glucans), triterpenoids (ganoderic acids), peptides, sterols, and phenolic compounds, exhibit potent anti-aging properties through diverse mechanisms. Polysaccharides enhance antioxidant defences, reduce free radical accumulation, and modulate immune responses, thereby delaying immunosenescence. Triterpenoids protect mitochondria, suppress pro-inflammatory signaling, and prevent apoptosis, contributing to neuroprotective and cardioprotective effects. Peptides and sterols further alleviate oxidative DNA damage, promote tissue repair, and maintain membrane stability, while phenolic compounds regulate cellular redox pathways and delay senescence. Experimental studies in animal and cell models have demonstrated that *Ganoderma lucidum* extracts improve lifespan, cognitive function, and stress resistance, supporting its role as a natural geroprotective agent. Moreover, its multi-targeted activity makes it a promising candidate in preventing age-related diseases, including neurodegeneration, cardiovascular disorders, and chronic kidney disease.

Keywords: *Ganoderma lucidum*; Aging; DNA Damage; Bioactive Compounds

1. Introduction

Aging is a complex biological process characterized by gradual decline in physiological function, increased oxidative stress, and susceptibility to chronic diseases such as neurodegeneration, cardiovascular disorders, diabetes, and cancer. Oxidative damage, mitochondrial dysfunction, immune system dysregulation, and chronic inflammation are considered major hallmarks of aging [1]. In recent decades, attention has shifted toward natural products as safe anti-aging agents, and medicinal mushrooms, particularly *Ganoderma lucidum* have gained global recognition for their health-promoting properties.

Mushrooms have long been valued not only as a food source but also for their medicinal potential in various traditional healing systems. *Ganoderma lucidum* is recognized by different names worldwide such as “Reishi” in Japan, meaning spiritual potency; “Lingzhi” in China, meaning divine mushroom; and “Youngzhi” in Korea, meaning mushroom of immortality [2]. It has been an integral part of traditional medicine in these countries for centuries, valued for its health-promoting properties. In comparison, scientific investigation into its benefits in the West has emerged only over the past three decades [3]. It has been revered for over two millennia as a symbol of health, longevity, and spiritual potency [4,5,6]. Ancient Chinese medical texts describe it as a “superior herb” capable of promoting vitality, enhancing resistance to illness, and extending lifespan [7,8]. Over recent decades, *G. lucidum* has transitioned from being a cornerstone of traditional Eastern medicine to a globally recognized medicinal resource, attracting significant interest from researchers, clinicians, and the pharmaceutical industry [9,10].

* Corresponding author: Newton Paul.

Ganoderma lucidum contains around 400 bioactive compounds mainly polysaccharides, triterpenoids, proteins, enzymes, steroids, sterols, nucleotides, fatty acids, vitamins, and minerals responsible for a wide range of therapeutic effects [11,12,13,14,15]. Among these, *Ganoderma lucidum* polysaccharides (GLPS) are considered the most significant component in anti-aging effects due to their powerful antioxidant, immunomodulatory, neuroprotective, anti-inflammatory, and anti-apoptotic activities. GLPS influence multiple cellular pathways associated with longevity and have been tested in models of aging and age-related diseases. Its pharmacological activities include immune modulation, anticancer, antidiabetic, antioxidant, anti-inflammatory, antimicrobial, hepatoprotective, and many other disease-preventing properties [9,16,17,18,19]. Modern scientific investigations have elucidated multiple mechanisms underlying these properties, including modulation of immune cell activity, suppression of pro-inflammatory mediators, scavenging of reactive oxygen species, regulation of apoptotic pathways in cancer cells, and improvement of liver function. These effects are directly related to delaying the aging process and preventing age-associated diseases.

The mushroom's versatility is reflected in its availability in various forms, including dried fruiting bodies, powdered extracts, capsules, teas, and functional food products. Nevertheless, challenges remain in standardizing its bioactive components, improving bioavailability, and validating its efficacy through large-scale, well-controlled clinical trials [10]. *Ganoderma lucidum* is a rare wood-degrading basidiomycete that is now cultivated artificially on logs, sawdust, or in bioreactors. Its main pharmacologically active compounds are triterpenoids, which exhibit hepatoprotective, antihypertensive, hypocholesterolemic, anti-tumor, and anti-angiogenic effects, and polysaccharides particularly β -D-glucans which have anti-tumor, immunomodulatory, antioxidant, and cell-protective properties [16]. Interest in *G. lucidum* has further intensified due to its potential applications in managing lifestyle-related disorders, supporting cancer therapy, and enhancing overall wellness. *Ganoderma*, the dried fruiting body of *Ganoderma lucidum* or *Ganoderma sinense*, has been used in traditional Chinese medicine for over 2000 years to treat respiratory issues, promote nourishment, and enhance strength. It contains over 470 natural compounds mainly terpenoids, steroids, alkaloids, and phenols with proven antitumor, anti-inflammatory, hypoglycemic, hypolipidemic, and immunomodulatory effects. Clinically, it is used for conditions such as diabetic nephropathy and malignant tumors, offering high safety and minimal side effects [18]. Nanotechnology has been applied to improve the extraction, purification, and stability of *Ganoderma lucidum* bioactive compounds, while nanoformulations enhance their therapeutic potential particularly in targeting cancer cells, inducing apoptosis, and inhibiting tumor growth. The integration of *G. lucidum* with nanotechnology offers promising biomedical applications, though further research and clinical trials are needed to optimize formulations and validate mechanisms for cancer therapy and other health benefits [20].

1.1. Taxonomy

Ganoderma is a genus of polypore fungi belonging to the family Ganodermataceae, renowned for its medicinal properties and traditional use in Asian herbal medicine, particularly *Ganoderma lucidum*. *Ganoderma* species grow naturally on decaying wood, stumps, or tree trunks (Figure1). They are widely distributed worldwide, from tropical and subtropical forests to temperate regions. Southeast Asia, Africa, and South America host the greatest diversity, while certain species are also frequently observed in North America and Europe (Table 1). Globally, scientists have identified over 430–490 species of *Ganoderma* [21]. The exact number is still debated because some species are morphologically similar and require molecular DNA sequencing for precise identification [22,23,24]

- Kingdom: Fungi
- Phylum: Basidiomycota
- Class: Agaricomycetes
- Order: Polyporales
- Family: Ganodermataceae
- Genus: *Ganoderma* P. Karst. (1881) [25]

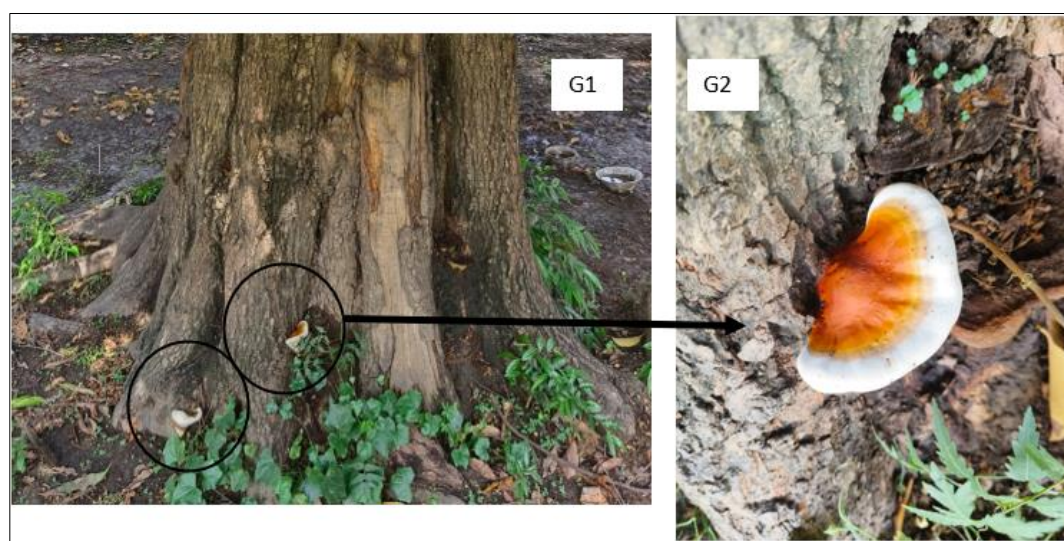


Figure 1 Images of *Ganoderma*; G1: Mushroom grow on the tree; G2: Mushroom enlarge view

Table 1 Six species of *Ganoderma* commonly used

Species	Country / Region of Origin
<i>Ganoderma lucidum</i>	Native to East Asia (China, Japan, Korea); also found in tropical/subtropical regions worldwide
<i>Ganoderma capense</i>	Distributed mainly in South Africa and some parts of southern China
<i>Ganoderma sinense</i>	Endemic to China, especially southern provinces; also in Taiwan and Japan
<i>Ganoderma applanatum</i>	Cosmopolitan – widely distributed in Europe, North America, Asia, and Africa
<i>Ganoderma tsugae</i>	Native to North America (particularly northeastern USA and Canada) and East Asia on hemlock trees
<i>Ganoderma boinense</i>	Originally described from Brazil; occurs in other parts of South America

1.2. Bioactive compounds of *Ganoderma lucidum* in aging

Aging is influenced by genetic and environmental factors, and polysaccharides from plants play an important anti-aging role through their antioxidant activity and regulation of key aging-related molecules like mTOR, AMPK, and p53. Although their direct absorption into the blood remains controversial, polysaccharides can modulate gut microbiota, leading to metabolic changes that impact aging pathways. Thus, gut microbiota-mediated mechanisms are a promising direction for studying the anti-aging effects of polysaccharides [26].

2. Polysaccharides

The bioactive compounds of *Ganoderma lucidum* polysaccharides, triterpenoids, peptides, sterols, and phenolics work synergistically to delay aging by reducing oxidative stress; enhancing immunity; inhibiting inflammation; protecting mitochondria and regulating gene pathways (Nrf2, NF- κ B, SIRT1, FOXO). Food-derived polysaccharides act as antioxidants by scavenging free radicals, regulating oxidative damage, and modulating stress-related pathways. They show neuroprotective effects in neurodegenerative disease models by reducing oxidative, inflammatory, and proteotoxic stresses. While most research focuses on plant and fungi derived polysaccharides, marine sources remain underexplored. Additionally, their immunomodulatory properties may also contribute to neuroprotection [27]. Polysaccharides (especially β -D-glucans) have a strong antioxidant effect by scavenging free radicals and boosting antioxidant enzymes (SOD, CAT, GSH-Px), modulating the immune system, preventing age-related immune decline (immunosenescence) and reducing chronic inflammation (“inflammaging”). It delays cellular senescence, improves memory and learning, and enhances resistance to oxidative stress [28,29]. Polysaccharides exert anti-aging effects

across multiple experimental models such as *C. elegans*, *Drosophila*, and mice via mechanisms including improved antioxidant capacity, immune regulation, reduced inflammation, and telomere protection [29].

Dietary β -1,3-glucan supplementation in the short-lived fish *Nothobranchius guentheri* delayed cellular senescence markers (e.g., SA- β -Gal, lipofuscin) and extended both median and maximum lifespan, demonstrating antioxidative and anti-senescence actions. The oxidative stress hypothesis of aging was tested in the fish *Nothobranchius guentheri*, where dietary β -1,3-glucans prolonged lifespan, reduced age-related biomarkers, lowered oxidative damage, and enhanced antioxidant enzyme activity. These findings suggest that β -1,3-glucans have significant anti-aging and antioxidant potential, indicating possible applications in elderly health care [30].

3. Triterpenoids

3.1. Ganoderic Acid D (GA-D)

Ganoderic acid D (GA-D), a triterpenoid from *Ganoderma lucidum*, shows strong anti-senescence effects in stem cells by activating the PERK/NRF2 pathway, highlighting its potential for developing anti-aging supplements and preventing aging-related diseases. Ganoderic Acid D has been shown to prevent oxidative stress-induced senescence in human amniotic mesenchymal stem cells (hAMSCs). It activates the PERK/NRF2 signaling pathway, reducing key senescence markers like SA- β -gal, p16, p21, and ROS [31].

Ganoderic acid D (GA-D) protects human amniotic mesenchymal stem cells from senescence by targeting 14-3-3 ϵ and activating the CaM/CaMKII/Nrf2 signaling pathway [32]. Stem cell senescence plays a key role in aging, and delaying it may help combat age-related diseases. Knockdown of the 14-3-3 ϵ gene reversed GA-D's protective effects, while its overexpression enhanced them. *In vivo*, GA-D also reduced oxidative stress, cell cycle arrest, and senescence markers in aging mouse models, while improving stem cell viability and differentiation. These findings suggest that GA-D delays stem cell senescence and exerts anti-aging effects via the 14-3-3 ϵ -CaM/CaMKII/Nrf2 pathway.

3.2. Ganoderic Acid A (GAA)

In aging mice, long-term administration of *Ganoderma* triterpenoids (TGL) mitigated signs of aging such as cataracts, hair loss, skin sagging, liver and spleen dysfunction, and kidney β -galactosidase activity. TGL also improved brain function. Research suggests that GAA may be a key active component. Mechanisms appear to involve regulated sphingolipid metabolism, telomere preservation, and enhanced autophagy. Long-term administration of *Ganoderma lucidum* triterpenoids (TGL) alleviates age-related physiological decline in mice, improving brain function, organ health, and pathological features such as cataracts, hair loss, and skin relaxation. In Alzheimer's models, ganoderic acid A was identified as a key constituent, potentially acting through sphingolipid metabolism regulation, telomere length extension, and enhanced autophagy to support healthy aging and neuroprotection [33]. GAA and related ganoderic acids are already being utilized in anti-aging skincare products. They promote keratinocyte proliferation, activate cell cycle proteins, enhance skin migration and remodeling, and exhibit antioxidant properties [34].

Alzheimer's disease (AD) is a progressive neurodegenerative condition that severely impacts memory and cognitive abilities in older adults. Although considerable research has been conducted, an absolute cure has not yet been discovered. Current treatments may help manage symptoms or slow progression, but they do not eliminate the disease. Continued investigation is essential to uncover effective therapeutic strategies and ultimately develop a permanent solution for patients. Existing treatments provide only limited relief, highlighting the urgent need for novel therapeutic strategies to effectively manage or prevent AD. Mushrooms, rich in bioactive compounds like polysaccharides, terpenoids, phenols, ergothioneine, and erinacine, show promising neuroprotective potential for AD prevention and management. Their compounds may act through mechanisms such as crossing the blood brain barrier, influencing neurotransmitters, and regulating brain function. While mushrooms are promising functional foods for reducing AD risk, more large-scale studies and clinical trials are needed to determine optimal dosage, bioavailability, interactions, and efficacy. Ganoderic acid A has an influential role in Alzheimer's Disease & Cognitive Protection [35]. Alzheimer's disease (AD) is a major neurodegenerative disorder and a leading cause of death in the elderly, yet effective treatments remain limited. The accumulation of amyloid- β (A β) peptides and their aggregates plays a key role in disease progression, but current drugs targeting A β , including γ -secretase inhibitors and monoclonal antibodies, have failed in clinical trials due to poor clearance of A β plaques. Research on *Ganoderma lucidum* (Reishi mushroom) reveals that its bioactive components, particularly polysaccharides and triterpenoids, show promise for neurological disorders. Specifically, ganoderic acid A (GAA) enhances autophagy, promotes A β 42 clearance by microglia, and improves memory and cognition in AD mouse models. GAA achieves this by activating Axl and Pak1 signaling pathways, which upregulate autophagy-related genes and facilitate lysosomal degradation of A β . These findings suggest that GAA from *G. lucidum*

spore powder could be a potential therapeutic for AD by targeting impaired A β clearance and restoring neuronal health [35].

In Alzheimer's Disease cell models (e.g., A β 25-35 treated HT22), GAA reduces cell senescence via JAK–PI3K–AKT–mTOR pathway modulation. It also enhances autophagy and A β clearance through Axl/Pak1 signaling, improving cognition in Alzheimer's Disease mice. In D-galactose mice (aging model), GAA modulated the Th17/Treg axis to reduce neuroinflammation and mitochondrial dysfunction [34,36,37].

4. Lucidones (Lucidone A)

Ganoderma lucidum has long been used to enhance brain function, and modern studies show its potential in treating neurodegenerative diseases (NDDs) through multiple mechanisms such as anti-AChE activity, reducing β -amyloid toxicity, stimulating neurite outgrowth, alleviating mitochondrial and ER stress, and providing antioxidant and anti-inflammatory effects. It offers advantages like low toxicity, multi-target action, and possible synergy with conventional therapies. However, challenges remain, including a lack of studies with pure compounds, variability in preparations, unclear mechanisms, limited clinical trials, and insufficient standardized treatments. Preliminary evidence, including a small case study in Parkinson's disease and a controlled trial in neurasthenia, suggests possible benefits, but more rigorous clinical research is needed.

Ganoderma lucidum shows significant potential as a supportive therapy for neurodegenerative diseases (NDDs). By integrating ancient medicinal wisdom with modern scientific research, it offers novel insights and possibilities for developing effective treatment strategies, highlighting its role as a bridge between traditional healing and contemporary therapeutic approaches [37]. *Ganoderma lucidum* alcohol extracts show anti-aging and neuroprotective effects in Alzheimer's disease (AD) models by modulating DNA methylation. Key active compounds, including ganoderic acids (Mk, C6) and lucidone A, up-regulated methylation regulators such as Histone H3, DNMT3A, and DNMT3B, improved memory and learning, reduced neuronal apoptosis and brain atrophy, and lowered A β 1-42 levels. These findings suggest *G. lucidum* extracts can delay Alzheimer's Disease progression through epigenetic regulation [38].

5. Peptides and Proteins

A major peptide component, known as *Ganoderma lucidum* peptide (GLP), demonstrates highly potent antioxidant activity across multiple models. It inhibits lipid peroxidation in systems like soybean oil and lard, outperforming some synthetic antioxidants. GLP scavenges hydroxyl and superoxide radicals effectively (with IC₅₀ values around 25 μ g/mL). It reduces malondialdehyde levels and prevents haemolysis in rat tissues, underscoring its protective potential against oxidative damage linked to aging. *Ganoderma lucidum* peptide (GLP) is identified as the major antioxidant component of the mushroom, surpassing polysaccharides and phenolics [39]. GLP demonstrated strong antioxidant effects in various systems by inhibiting lipid peroxidation, blocking soybean lipoxygenase, scavenging hydroxyl and superoxide radicals, and reducing malondialdehyde levels. It also protected rat liver tissue, mitochondrial membranes, and red blood cells from oxidative damage. The results indicate that GLP significantly contributes to the antioxidant effects of *G. lucidum*, primarily by scavenging free radicals, chelating metals, and preventing lipid peroxidation [40,41].

Ganoderma lucidum Polysaccharide–Peptide (GLPP), a complex of polysaccharides with ~5% peptide content, combines antioxidant and immunomodulatory functions. Chemotherapy, particularly with cyclophosphamide (CP), causes severe testicular damage and male infertility by inducing oxidative stress, apoptosis, and disruption of spermatogenesis. *Ganoderma lucidum* polysaccharide peptide (GLPP), a natural antioxidant, was tested in a CP-treated mouse model. Results showed that GLPP protected germ cells at different spermatogenic stages by reducing oxidative stress, restoring antioxidant enzyme activities (SOD, GSH, CAT), lowering lipid peroxidation (MDA), and upregulating Nrf2/HO-1 signalling. GLPP also improved sperm count, motility, morphology, and sex hormone levels, while maintaining testis and epididymis indexes without toxicity. In the study interaction patterns of Bax and Bak with anti-apoptotic Bcl-2 family proteins were studied. Bcl-B specifically binds Bax and inhibits Bax-induced apoptosis, while Mcl-1 selectively binds Bak and blocks Bak-induced apoptosis. In contrast, other family members (Bcl-2, Bcl-XL, Bcl-W, Bfl-1) interact with both Bax and Bak, suppressing apoptosis from either. The results demonstrate the distinct selectivity of Bcl-B and Mcl-1, providing valuable understanding of how Bax- and Bak-mediated cell death is regulated [42]. This modulation of the Bax/Bcl-2 ratio highlights GLPP's protective role in maintaining cellular survival mechanisms and suggests its potential therapeutic significance against CP-associated cytotoxicity and oxidative stress damage [43]. Overall, GLPP showed strong protective effects against chemotherapy-induced spermatogenic injury, highlighting its potential as a safe therapeutic agent for male infertility linked to oxidative stress. This suggests systemic anti-aging benefits via oxidative stress reduction and cellular protection [44].

6. Phenolic Compounds

Ganoderma lucidum contains phenolics, including Phenolic acids (gallic acid, caffeic acid, ferulic acid, benzoic acid, p-hydroxybenzoic acid, p-coumaric acid, cinnamic acid derivatives) and Flavonoids (quercetin, kaempferol, hesperetin, naringenin, myricetin, morin, biochanin, formononetin) [45,46]. These compounds exhibit strong free radical scavenging, reduce oxidative stress, inhibit lipid peroxidation, and support antioxidant enzyme activity, key factors in slowing cellular aging. Ethanol extracts rich in phenolics boost mitochondrial antioxidant enzymes (SOD, CAT, GSH-Px), support GSH levels, and enhance radical scavenging activities like DPPH and ABTS, thereby promoting cellular health and longevity. Aqueous extracts, containing phenolics, may help protect neuronal cells through inhibition of oxidative stress-related pathways, including JNK and p38 signaling [47,48].

Ganoderma, traditionally used as an elixir for longevity, shows potential anti-aging effects, though its exact role in humans remains unclear. Aging is linked to immune dysregulation, oxidative stress, and free radical production. Studies suggest *Ganoderma* may extend lifespan by reducing ROS production, lipid peroxidation, and advanced oxidation protein products, while enhancing mitochondrial electron transport complexes, antioxidant enzymes (SOD, CAT, GSH, GSH-Px), and radical scavenging activities (DPPH, ABTS). It also exhibits strong immunomodulatory and antioxidant effects, including ferric reducing antioxidant power. Despite these promising findings, further studies are needed to clarify its mechanisms and clinical applications in aging [48].

7. Sterols

Sterols in *Ganoderma lucidum*, including ergosterol and ergosta-5,7,22-trien-3 β -ol, are vital components of the fungal cell membrane, contributing to membrane fluidity and serving as precursors to vitamin D₂. Ergosterol is particularly abundant and often used as a quality marker for *G. lucidum* extracts. Its stability and conversion into vitamin D₂ under UV exposure have been widely studied. Typically, sterols are extracted using nonpolar solvents such as hexane or supercritical CO₂, ensuring effective isolation. These sterols not only support fungal physiology but also enhance the nutritional and medicinal value of *G. lucidum*, linking it to health-promoting and therapeutic applications [49,50,51]. Ergosterol and ergosta-5,7,22-trien-3 β -ol, among others, are integral to fungal cell membranes and serve as precursors to vitamin D₂. Ergosterol, a key sterol in edible mushrooms, serves as a provitamin and precursor of vitamin D₂ when exposed to UV radiation. Beyond its nutritional value, ergosterol exhibits diverse pharmacological effects, including antimicrobial, antioxidant, anticancer, antidiabetic, and anti-neurodegenerative activities, making mushrooms an important functional food source [51].

8. Discussion

Aging is a complex biological phenomenon shaped by the interplay of genetic factors and environmental influences. The bioactive compounds of *Ganoderma lucidum* (GL), particularly polysaccharides and triterpenoids, have been studied extensively for their ability to counteract hallmarks of aging, including oxidative stress, inflammation, mitochondrial dysfunction, and cellular senescence. The GL acts through multiple pathways to support healthy aging, positioning it as a promising candidate for therapeutic and functional food applications.

Polysaccharides, especially β -D-glucans, are among the most important anti-aging constituents of GL. They exert strong antioxidant effects by scavenging free radicals and enhancing endogenous defence systems such as superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GSH-Px). These activities mitigate oxidative damage, which is central to the “free radical theory of aging.” Evidence from animal models—including *C. elegans*, *Drosophila*, and mice—demonstrates that GL polysaccharides improve antioxidant capacity, reduce chronic inflammation, protect telomeres, and delay cellular senescence [29]. In short-lived fish models, β -1,3-glucans extended both median and maximum lifespan, supporting their role as potent anti-senescence agents [30]. Beyond direct antioxidantation, polysaccharides modulate the gut microbiota, indirectly influencing systemic metabolism and aging pathways such as mTOR, AMPK, and p53 [26]. This highlights the importance of host-microbiome interactions in dietary anti-aging strategies.

In addition to polysaccharides, GL triterpenoids exhibit remarkable anti-aging potential. Ganoderic acid D (GA-D) delays stem cell senescence by activating the PERK/NRF2 and CaM/CaMKII/Nrf2 signaling axes, reducing ROS accumulation and senescence markers (p16, p21, SA- β -gal) in human amniotic mesenchymal stem cells [31,32]. Since stem cell exhaustion is a critical driver of aging, these findings are particularly significant. Likewise, Ganoderic acid A (GAA) has demonstrated neuroprotective effects in aging and Alzheimer’s disease (AD) models. By enhancing autophagy and facilitating amyloid- β clearance, GAA improves cognition and mitigates neurodegeneration [35,36]. The ability of

triterpenoids to regulate telomere preservation, sphingolipid metabolism, and Th17/Treg immune balance further suggests their potential in systemic anti-aging interventions [33].

Other GL bioactives, including peptides, proteins, phenolics, and sterols, complement these effects. *Ganoderma lucidum* peptide (GLP) is a highly potent antioxidant, capable of preventing lipid peroxidation and mitochondrial damage [39]. Phenolics such as quercetin, gallic acid, and ferulic acid contribute additional antioxidant and neuroprotective activities by modulating stress-related pathways (e.g., NF- κ B, Nrf2) [48]. Sterols like ergosterol serve not only as vitamin D₂ precursors but also exhibit anti-inflammatory and neuroprotective activities, expanding the health-promoting potential of GL [51]. Collectively, GL functions through multi-target mechanisms attenuating oxidative damage, improving mitochondrial function, regulating stress responses, and protecting neural and stem cell integrity. Such pleiotropic actions align well with the multifactorial nature of aging.

9. Conclusion

Ganoderma lucidum contains a wide range of bioactive compounds particularly polysaccharides, triterpenoids, peptides, phenolics, and sterols that collectively exert significant anti-aging effects. Polysaccharides, particularly β -D-glucans, along with triterpenoids like ganoderic acids A and D, are considered the primary bioactive components responsible for their effects. Polysaccharides primarily act through antioxidant activity, immune regulation, telomere protection, and modulation of gut microbiota, thereby alleviating oxidative stress, immunosenescence, and chronic inflammation. Triterpenoids, on the other hand, show strong neuroprotective and anti-senescence effects, especially by activating NRF2 and autophagy-related pathways, which are crucial for healthy aging and cognitive preservation. Other compounds, such as peptides (GLP, GLPP), phenolics, and sterols, further enhance antioxidant defense, cellular protection, and neuroprotection, highlighting their synergistic effects. The synergistic interaction of these bioactive molecules highlights the holistic anti-aging profile of *G. lucidum*. By simultaneously targeting multiple hallmarks of aging oxidative stress, chronic inflammation ("inflammaging"), immune decline, telomere shortening, and impaired autophagy the mushroom provides a multi-layered protective effect that surpasses the efficacy of many single-compound interventions. In conclusion, *Ganoderma lucidum* represents a valuable natural resource for anti-aging strategies. With further translational research, GL has the potential to serve as both a functional food and a pharmacological agent in combating age-related decline.

Compliance with ethical standards

Acknowledgments

The author is grateful to the Principal Dr. (Maj.) Neerja Masih, Isabella Thoburn College, Lucknow, Uttar Pradesh, India for her continuous encouragement and guidance.

Disclosure of conflict of interest

The Authors proclaim no conflict of interest.

References

- [1] López-Otín C, Blasco MA, Partridge L, Serrano M, Kroemer G. The hallmarks of aging. *Cell*. 2013 Jun 6;153(6):1194-217. doi: 10.1016/j.cell.2013.05.039. PMID: 23746838; PMCID: PMC3836174.
- [2] Paterson, R. R. M. *Ganoderma* – A therapeutic fungal biofactory. *Phytochemistry*, 2006, 67(18), 1985-2001.
- [3] Cai, Q.; Li, Y.; Pei, G. Polysaccharides from *Ganoderma lucidum* attenuates microglia-mediated neuroinflammation and modulate microglial phagocytosis and behavioural response. *J. Neuroinflamm.* 2017, 14, 63–76.
- [4] Sliva, D. Cellular and physiological effects of *Ganoderma lucidum* (Reishi). *Mini Rev. Med. Chem.* 2004, 4, 873–879.
- [5] Li, H.J.; He, Y.L.; Zhang, D.H.; Yue, T.H.; Jiang, L.X.; Li, N.; Xu, J.W. Enhancement of ganoderic acid production by constitutively expressing *Vitreoscilla* hemoglobin gene in *Ganoderma lucidum*. *J. Biotechnol.* 2016, 227, 35–40.
- [6] Soccol, C.R.; Bissoqui, L.Y.; Rodrigues, C.; Rubel, R.; Sella, S.R.; Leifa, F.; De Souza Vandenberghe, L.P.; Soccol, V.T. Pharmacological Properties of Biocompounds from Spores of the Lingzhi or Reishi Medicinal Mushroom *Ganoderma lucidum* (Agaricomycetes): A Review. *Int. J. Med. Mushrooms* 2016, 18, 757–767.

- [7] Zheng, W. Drug discovery from fungal metabolites: A review of the papers in this monographical issue of Mycosystema concerned with the natural resources, problems and strategies. Mycosystema 2011, 30, 151–157.
- [8] Ahmad, M.F.; Ahmad, F.A.; Azad, Z.R.; Azaz, A.; Alam, M.I.; Ansari, J.A.; Panda, B.P. Edible mushrooms as health promoting agent. Adv. Sci. Focus 2013, 1, 189–196.
- [9] Ahmad, M.F. *Ganoderma lucidum*: Persuasive biologically active constituents and their health endorsement. Biomed. Pharmacother. 2018, 107, 507–519.
- [10] Cadar, E.; Negreanu-Pirjol, T.; Pascale, C.; Sirbu, R.; Prasacu, I.; Negreanu-Pirjol, B.-S.; Tomescu, C.L.; Ionescu, A.-M. Natural Bio-Compounds from *Ganoderma lucidum* and Their Beneficial Biological Actions for Anticancer Application: A Review. Antioxidants 2023, 12, 1907. <https://doi.org/10.3390/antiox12111907>
- [11] Smania, E.F.A.; Monache, F.D.; Smania, A., Jr.; Yunes, R.A.; Cuneo, R.S. Antifungal activity of sterols and triterpenes isolated from *Ganoderma annulare*. Fitoterapia 2003, 74, 375–377.
- [12] Sanodiya, B.S.; Thakur, G.S.; Baghel, R.K.; Prasad, G.B.; Bisen, P.S. *Ganoderma lucidum*: A potent pharmacological macrofungus. Curr. Pharm. Biotechnol. 2009, 10, 717–742.
- [13] Ferreira, I.C.; Vaz, J.; Vasconcelos, M.H.; Martins, A. Compounds from wild mushrooms with antitumor potential. Anticancer. Agents Med. Chem. 2010, 10, 424–436.
- [14] El Sheikha, A.F. Nutritional Profile and Health Benefits of *Ganoderma lucidum* “Lingzhi, Reishi, or Mannentake” as Functional Foods: Current Scenario and Future Perspectives. Foods 2022, 11, 1030.
- [15] Ekiz, E.; Oz, E.; Abd El-Aty, A.M.; Proestos, C.; Brennan, C.; Zeng, M.; Tomasevic, I.; Elobeid, T.; Çadırcı, K.; Bayrak, M.; et al. Exploring the Potential Medicinal Benefits of *Ganoderma lucidum*: From Metabolic Disorders to Coronavirus Infections. Foods 2023, 12, 1512. <https://doi.org/10.3390/foods12071512>
- [16] Boh B, Berovic M, Zhang J, Zhi-Bin L. *Ganoderma lucidum* and its pharmaceutically active compounds. Biotechnol Annu Rev. 2007;13:265-301. doi: 10.1016/S1387-2656(07)13010-6. PMID: 17875480.
- [17] Cör Andrejč D, Knez Ž, Knez Marevci M. Antioxidant, antibacterial, antitumor, antifungal, antiviral, anti-inflammatory, and neuro-protective activity of *Ganoderma lucidum*: An overview. Front Pharmacol. 2022 Jul 22;13:934982. doi: 10.3389/fphar.2022.934982. PMID: 35935849; PMCID: PMC9353308.
- [18] Liu C, Song X, Li Y, Ding C, Li X, Dan L, Xu H, Zhang D. A Comprehensive Review on the Chemical Composition, Pharmacology and Clinical Applications of *Ganoderma*. Am J Chin Med. 2023;51(8):1983-2040. doi: 10.1142/S0192415X23500878. Epub 2023 Oct 31. PMID: 37903715.
- [19] Zhong, Y.; Tan, P.; Lin, H.; Zhang, D.; Chen, X.; Pang, J.; Mu, R. A Review of *Ganoderma lucidum* Polysaccharide: Preparations, Structures, Physicochemical Properties and Application. Foods 2024, 13, 2665. <https://doi.org/10.3390/foods13172665>
- [20] Eira A, Gonçalves MBS, Fongang YSF, Domingues C, Jarak I, Mascarenhas-Melo F, Figueiras A. Unlocking the Potential of *Ganoderma lucidum* (Curtis): Botanical Overview, Therapeutic Applications, and Nanotechnological Advances. Pharmaceutics. 2025 Mar 26;17(4):422. doi: 10.3390/pharmaceutics17040422. PMID: 40284417; PMCID: PMC12030363
- [21] Waill A. Elkhateeb, Ghoson M. Daba. The precious *Ganoderma* mushroom and plant diseases. Journal of Pharmacy and Drug Development. 2022; 1(1). DOI: 10.58489/2836-2322/005
- [22] Moncalvo, J.M., & Ryvarden, L. A nomenclatural study of the Ganodermataceae. Synopsis Fungorum, 1997; 11, 1–114.
- [23] Wasser, S.P. Medicinal mushrooms as a source of antitumor and immunomodulating polysaccharides. Applied Microbiology and Biotechnology, 2010; 89(1), 132–153.
- [24] Hapuarachchi K.K., Karunarathna S.C., Raspé O., De Silva K.H.W.L., Thawthong A., Wu X.L., Kakumyan P., Hyde K.D., Wen T.C. High diversity of *Ganoderma* and *Amauroderma* (Ganodermataceae, Polyporales) in Hainan Island, China. Mycosphere 2018; 9(5), 931–982, Doi 10.5943/mycosphere/9/5/1
- [25] Kirk, P.M., Cannon, P.F., Minter, D.W., & Stalpers, J.A. Dictionary of the Fungi, 2008; 10th ed. CABI.
- [26] Xu W, Han S, Huang M, Yin J, Yang F, Luo F. Antiaging Effects of Dietary Polysaccharides: Advance and Mechanisms. Oxid Med Cell Longev. 2022 Jul 12; 4362479. doi: 10.1155/2022/4362479. PMID: 35864870; PMCID: PMC9296321.

- [27] Li, H.; Ding, F.; Xiao, L.; Shi, R.; Wang, H.; Han, W.; Huang, Z. Food-Derived Antioxidant Polysaccharides and Their Pharmacological Potential in Neurodegenerative Diseases. *Nutrients* 2017, 9, 778. <https://doi.org/10.3390/nu9070778>
- [28] Bai L, Xu D, Zhou YM, Zhang YB, Zhang H, Chen YB, Cui YL. Antioxidant Activities of Natural Polysaccharides and Their Derivatives for Biomedical and Medicinal Applications. *Antioxidants (Basel)*. 2022 Dec 19;11(12):2491. doi: 10.3390/antiox11122491. PMID: 36552700; PMCID: PMC9774958.
- [29] Guo, X.; Luo, J.; Qi, J.; Zhao, X.; An, P.; Luo, Y.; Wang, G. The Role and Mechanism of Polysaccharides in Anti-Aging. *Nutrients* 2022, 14, 5330. <https://doi.org/10.3390/nu14245330>
- [30] Song, Lili & Zhou, Yang & Ni, Shousheng & Wang, Xia & Yuan, Jiangshui & Zhang, Yu & Zhang, Shicui. Dietary Intake of β -Glucans Can Prolong Lifespan and Exert an Antioxidant Action of Aged Fish *Nothobranchius guentheri*. *Rejuvenation Research*. 2019; 23. 10.1089/rej.2019.2223.
- [31] Xu Y, Yuan H, Luo Y, Zhao YJ, Xiao JH. Ganoderic Acid D Protects Human Amniotic Mesenchymal Stem Cells against Oxidative Stress-Induced Senescence through the PERK/NRF2 Signaling Pathway. *Oxid Med Cell Longev*. 2020 Jul 27;2020:8291413. doi: 10.1155/2020/8291413. PMID: 32774686; PMCID: PMC7407022.
- [32] Yuan H, Xu Y, Luo Y, Zhang JR, Zhu XX, Xiao JH. Ganoderic acid D prevents oxidative stress-induced senescence by targeting 14-3-3 ϵ to activate CaM/CaMKII/NRF2 signaling pathway in mesenchymal stem cells. *Aging Cell*. 2022 Sep;21(9):e13686. doi: 10.1111/ace1.13686. Epub 2022 Aug 5. Erratum in: *Aging Cell*. 2022 Dec;21(12):e13732. doi: 10.1111/ace1.13732. PMID: 35929187; PMCID: PMC9470892.
- [33] Zeng M, Qi L, Guo Y, Zhu X, Tang X, Yong T, Xie Y, Wu Q, Zhang M, Chen D. Long-Term Administration of Triterpenoids from *Ganoderma lucidum* Mitigates Age-Associated Brain Physiological Dne via Regulating Sphingolipid Metabolism and Enhancing Autophagy in Mice. *Front Aging Neurosci*. 2021 May 6;13:628860. doi: 10.3389/fnagi.2021.628860. PMID: 34025387; PMCID: PMC8134542.
- [34] Wang S, Wang L, Shangguan J, Jiang A, Ren A. Research Progress on the Biological Activity of Ganoderic Acids in *Ganoderma lucidum* over the Last Five Years. *Life (Basel)*. 2024 Oct 21;14(10):1339. doi: 10.3390/life14101339. PMID: 39459639; PMCID: PMC11509451.
- [35] Qi LF, Liu S, Liu YC, Li P, Xu X. Ganoderic Acid A Promotes Amyloid- β Clearance (In Vitro) and Ameliorates Cognitive Deficiency in Alzheimer's Disease (Mouse Model) through Autophagy Induced by Activating Axl. *Int J Mol Sci*. 2021 May 24;22(11):5559. doi: 10.3390/ijms22115559. PMID: 34074054; PMCID: PMC8197357.
- [36] Li N, Li H, Liu Z, Feng G, Shi C, Wu Y. Unveiling the Therapeutic Potentials of Mushroom Bioactive Compounds in Alzheimer's Disease. *Foods*. 2023 Aug 7;12(15):2972. doi: 10.3390/foods12152972. PMID: 37569241; PMCID: PMC10419195.
- [37] Lian, W.; Yang, X.; Duan, Q.; Li, J.; Zhao, Y.; Yu, C.; He, T.; Sun, T.; Zhao, Y.; Wang, W. The Biological Activity of *Ganoderma lucidum* on Neurodegenerative Diseases: The Interplay between Different Active Compounds and the Pathological Hallmarks. *Molecules* 2024, 29, 2516. <https://doi.org/10.3390/molecules29112516>
- [38] Lai G., Guo Y., Chen D., Tang X., Shuai O., Yong T., Wang D., Xiao C., Zhou G., Xie Y., Yang B.B. and Wu Q. Alcohol Extracts from *Ganoderma lucidum* Delay the Progress of Alzheimer's Disease by Regulating DNA Methylation in Rodents. *Front. Pharmacol*. 2019; 10:272. doi: 10.3389/fphar.2019.00272
- [39] Jue Wang, Bin Cao, Haiping Zhao, Juan Feng. Emerging Roles of *Ganoderma lucidum* in Anti-Aging. *Aging and disease*. 2017, 8(6): 691-707 <https://doi.org/10.14336/AD.2017.0410>
- [40] Sun J, He H, Xie BJ. Novel antioxidant peptides from fermented mushroom *Ganoderma lucidum*. *J Agric Food Chem*. 2004 Oct 20;52(21):6646-52. doi: 10.1021/jf0495136. PMID: 15479035.
- [41] Shao Z, Zhang X, Cai J, Lu F. Glucagon-like peptide-1: a new potential regulator for mesenchymal stem cells in the treatment of type 2 diabetes mellitus and its complication. *Stem Cell Res Ther*. 2025 May 19;16(1):248. doi: 10.1186/s13287-025-04369-4. PMID: 40390070; PMCID: PMC12090506.
- [42] Zhai D, Jin C, Huang Z, Satterthwait AC, Reed JC. Differential regulation of Bax and Bak by anti-apoptotic Bcl-2 family proteins Bcl-B and Mcl-1. *J Biol Chem*. 2008 Apr 11;283(15):9580-6. doi: 10.1074/jbc.M708426200. Epub 2008 Jan 4. PMID: 18178565; PMCID: PMC2442277.
- [43] Toscano ECB, Vieira ÉLM, Portela ACDC, Reis JLJ, Caliarí MV, Giannetti AV, Gonçalves AP, Siqueira JM, Suemoto CK, Leite REP, Nitrini R, Teixeira AL, Rachid MA. Bcl-2/Bax ratio increase does not prevent apoptosis of glia and granular neurons in patients with temporal lobe epilepsy. *Neuropathology*. 2019 Oct;39(5):348-357. doi: 10.1111/neup.12592. Epub 2019 Aug 7. PMID: 31392787.

- [44] Zhang, H.; Li, N.; Zhang, Y.; Xu, Y.; Lu, F.; Lin, D.; Lin, S.; Li, M.; Yang, B. *Ganoderma lucidum* Polysaccharide Peptide Alleviates Cyclophosphamide-Induced Male Reproductive Injury by Reducing Oxidative Stress and Apoptosis. *Biomedicines* 2024, 12, 1632. <https://doi.org/10.3390/biomedicines12081632>
- [45] Veljović S, Veljović M, Nikićević N, Despotović S, Radulović S, Nikšić M, Filipović L. Chemical composition, antiproliferative and antioxidant activity of differently processed *Ganoderma lucidum* ethanol extracts. *J Food Sci Technol*. 2017 Apr;54(5):1312-1320. doi: 10.1007/s13197-017-2559-y. Epub 2017 Mar 10. PMID: 28416882; PMCID: PMC5380636.
- [46] Plosca MP, Chiş MS, Fărcaş AC, Păucean A. *Ganoderma lucidum*-from Ancient Remedies to Modern Applications: Chemistry, Benefits, and Safety. *Antioxidants (Basel)*. 2025 Apr 25;14(5):513. doi: 10.3390/antiox14050513. PMID: 40427395; PMCID: PMC12108272.
- [47] Wang J, Cao B, Zhao H, Feng J. Emerging Roles of *Ganoderma Lucidum* in Anti-Aging. *Aging Dis*. 2017 Dec 1;8(6):691-707. doi: 10.14336/AD.2017.0410. PMID: 29344411; PMCID: PMC5758346.
- [48] Pan Y, and Lin Z. Anti-aging Effect of *Ganoderma* (Lingzhi) with Health and Fitness. *Adv Exp Med Biol*. 2019; 1182:299-309. doi: 10.1007/978-981-32-9421-9_13. PMID: 31777025.
- [49] Obodai M, Mensah DL, Fernandes Â, Kortei NK, Dzomeku M, Teegarden M, Schwartz SJ, Barros L, Prempeh J, Takli RK, Ferreira IC. Chemical Characterization and Antioxidant Potential of Wild *Ganoderma* Species from Ghana. *Molecules*. 2017 Jan 25;22(2):196. doi: 10.3390/molecules22020196. PMID: 28125070; PMCID: PMC6155870.
- [50] Papoutsis K, Grasso S, Menon A., Brunton N.P., Lyng J G., Jacquier J C., Bhuyan D J.. 2020. Recovery of ergosterol and vitamin D2 from mushroom waste - Potential valorization by food and pharmaceutical industries, *Trends in Food Science & Technology*, Vol 99, 2020, 351-366,ISSN 0924-2244, <https://doi.org/10.1016/j.tifs.2020.03.005>.
- [51] Rangsinth P, Sharika R, Pattarachotananant N, Duangjan C, Wongwan C, Sillapachaiyaporn C, Nilkhet S, Wongsirojkul N, Prasansuklab A, Tencomnao T, Leung GP, Chuchawankul S. Potential Beneficial Effects and Pharmacological Properties of Ergosterol, a Common Bioactive Compound in Edible Mushrooms. *Foods*. 2023 Jun 29;12(13):2529. doi: 10.3390/foods12132529. PMID: 37444267; PMCID: PMC10340568.